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Summary of AWS Compute Services

Amazon Web Services (AWS) provides a variety of compute services that allow organizations to run applications and process data in the cloud without purchasing or maintaining physical servers. These services offer flexible computing resources and support four primary approaches: virtual machines, containers, serverless computing, and managed application platforms. Selecting the appropriate service depends on the application's design, its usage patterns, and the level of control required over the underlying infrastructure.

Virtual Machine-Based Computing

The foundational service for virtual machine computing is Amazon Elastic Compute Cloud (Amazon EC2). Amazon EC2 utilizes an Infrastructure as a Service (IaaS) model to provide resizable virtual machines, known as instances. Under this model, AWS supplies the physical hardware while customers maintain complete control over configuring the guest operating system, networking, and software applications.

To launch an EC2 instance, users must select an Amazon Machine Image (AMI), which is a template containing the required operating system and software configurations. Customers must also choose an instance type, which determines the specific combination of processor, memory, storage, and networking capacity. Instance types are grouped into families based on their purpose, such as General purpose, Compute optimized, Memory optimized, and Storage optimized. The configuration process also involves setting up Virtual Private Cloud (VPC) network settings, assigning IAM roles for service permissions, defining storage volumes, configuring security groups for network traffic, and selecting a key pair for secure connections.

Throughout its lifecycle, an EC2 instance moves from a pending state to a running state, where it becomes available for use. Instances can be temporarily stopped when computing capacity is unneeded and eventually terminated to permanently delete them. To optimize costs, AWS offers several pricing models depending on workload predictability: On-Demand Instances for flexible needs, Savings Plans and Reserved Instances for committed usage discounts, Spot Instances for discounted spare capacity, and Dedicated Hosts for specific compliance requirements.

Container-Based Computing

Containers provide a lightweight and portable way to deploy applications by packaging the application code together with its required libraries and dependencies. Unlike virtual machines, containers share the host operating system, allowing them to launch much

faster. Docker is a commonly used software platform for building and running these containerized applications via container images.

AWS provides multiple services for managing containerized workloads. Amazon Elastic Container Service (Amazon ECS) is a managed orchestration service that coordinates where containers run and maintains the desired capacity. Amazon Elastic Kubernetes Service (Amazon EKS) offers a managed environment for Kubernetes, an open-source system used to deploy and scale containerized applications. Both ECS and EKS can deploy containers using two primary compute options: Amazon EC2 instances for deep infrastructure control, or AWS Fargate, a serverless compute engine that removes the need to provision or manage underlying servers. To securely store, manage, and retrieve container images, AWS offers the Amazon Elastic Container Registry (Amazon ECR).

Serverless Computing

Serverless computing enables applications to run without requiring the customer to provision or manage any servers. AWS Lambda is a prominent serverless compute service where developers simply provide the code and configuration settings, organized into Lambda functions. AWS handles all the underlying infrastructure required to execute the code.

AWS Lambda relies on event-driven computing, meaning functions execute automatically in response to specific triggers. Examples of triggers include uploading an object to an Amazon S3 bucket, modifying data in an Amazon DynamoDB table, receiving an application request, or reaching a scheduled time. When triggered, Lambda automatically allocates resources, executes the code, and scales based on the volume of incoming requests. This model removes server administration burdens and utilizes a pay-for-use pricing structure based only on the compute resources consumed during execution.

Managed Application Platforms

For developers seeking to focus purely on coding without infrastructure management, AWS offers Elastic Beanstalk. AWS Elastic Beanstalk is a managed platform designed specifically for deploying and running web applications. Developers choose from supported platforms including Docker, Go, Java, .NET, Node.js, PHP, Python, and Ruby and upload their application code.

Once the code is uploaded, Elastic Beanstalk automatically creates an environment and handles the infrastructure provisioning, load balancing, capacity scaling, and health monitoring. While the deployment is vastly simplified, the underlying AWS resources remain accessible, granting developers the ability to adjust configurations if additional control is ever required.